

2010376103 CSE4261 Assignment-6

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1 Dataset

For this assignment, three images from the ImageNet dataset were used:

- **Fireboat** – originally predicted as crane.
- **Giant Panda** – originally predicted as giant panda.
- **Green Mamba** – originally predicted as green mamba.

Images were downloaded via direct URLs, resized to 224×224 pixels, and preprocessed using the standard MobileNet preprocessing pipeline.

2 Experimental Setup

2.1 Model

We used the pretrained **MobileNet** model from TensorFlow Keras with ImageNet weights. The model's weights were frozen during all experiments.

2.2 Adversarial Attack

Adversarial examples were generated using the **Fast Gradient Sign Method (FGSM)** with $\epsilon = 0.1$. The loss was defined as categorical cross-entropy between the predicted logits and the one-hot encoded true label.

2.3 Explainable AI Methods

Two XAI methods were applied to both original and adversarial images:

- **Integrated Gradients (IG)** – computed with respect to pre-softmax logits and a baseline of zeros.
- **Grad-CAM** – computed using the last convolutional layer. Two settings were compared:
 1. Using logits directly for Grad-CAM and pre-softmax gradients for IG.
 2. Using softmax layer for Grad-CAM and previous layer of softmax for IG.

3 Investigation of Adversarial Confusion using Grad-CAM and IG

For investigation, the original and adversarial images were fed into MobileNet. Figure 1 shows the top-3 predictions of original images and Figure 2 shows predictions after adversarial attack.

- Fireboat: originally predicted as crane (41.7%), after attack confidence increased to 91.2%.
- Giant Panda: confidence reduced from 99.9% to 92.6%.
- Green Mamba: prediction altered to television.

Adversarial attack caused significant confusion in the model. Attribution masks from IG (Figures 3, 4, 5) and heatmaps from Grad-CAM (Figure 6) highlight the regions responsible for confusion.

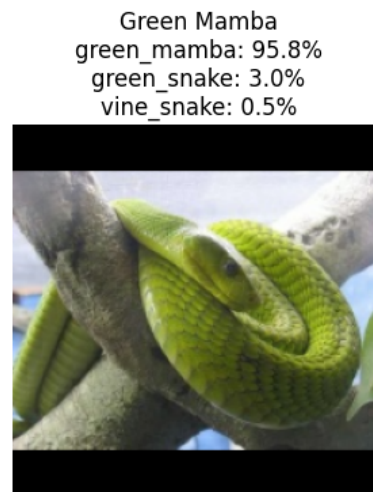


Figure 1: Top 3 prediction of original images

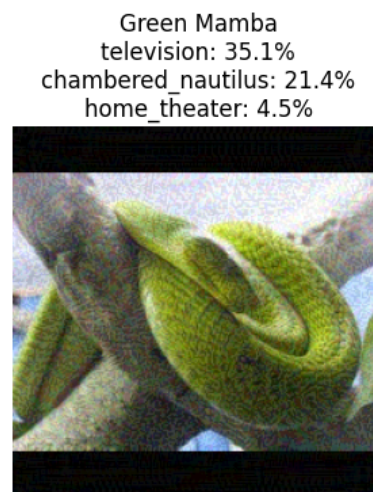
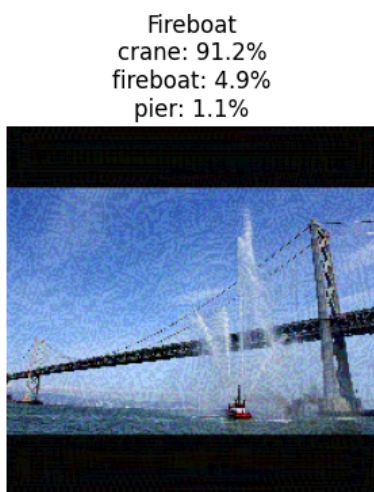


Figure 2: Top 3 prediction of adversarial images

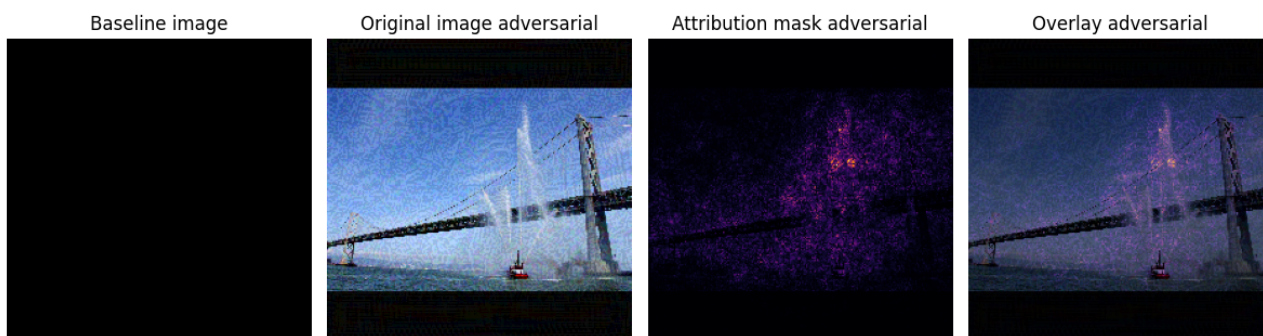


Figure 3: IG on adversarial Fireboat image

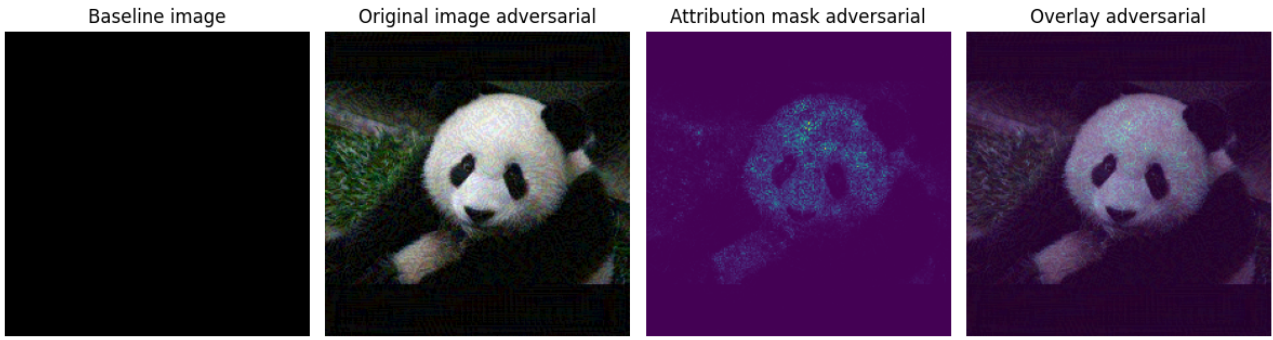


Figure 4: IG on adversarial Giant Panda image

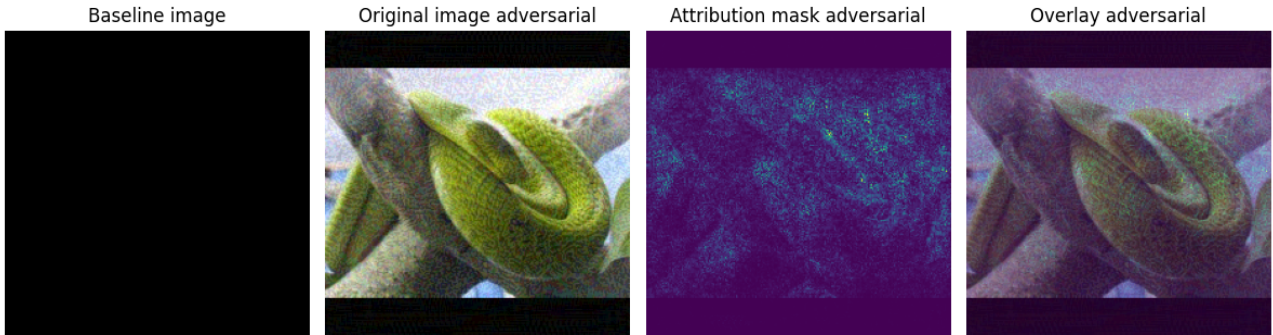


Figure 5: IG on adversarial Green Mamba image

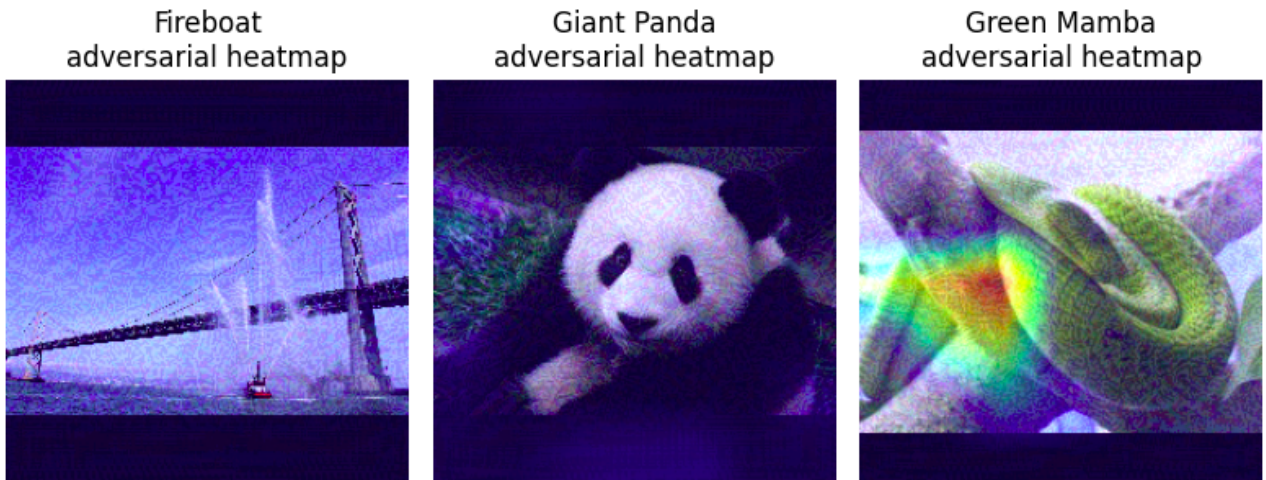


Figure 6: Grad-CAM on adversarial images

4 Comparison of Important Areas: Original vs Adversarial Images

Figures 7, 8, 9 show IG attribution masks for original and adversarial images, and Figure 10 shows Grad-CAM heatmaps.

- **Giant Panda:** minimal change in important regions.
- **Fireboat & Green Mamba:** important areas in adversarial images are more spread out compared to concentrated regions in original images.
- Heatmaps show that adversarial perturbation causes the model to attend to wider, less focused regions.

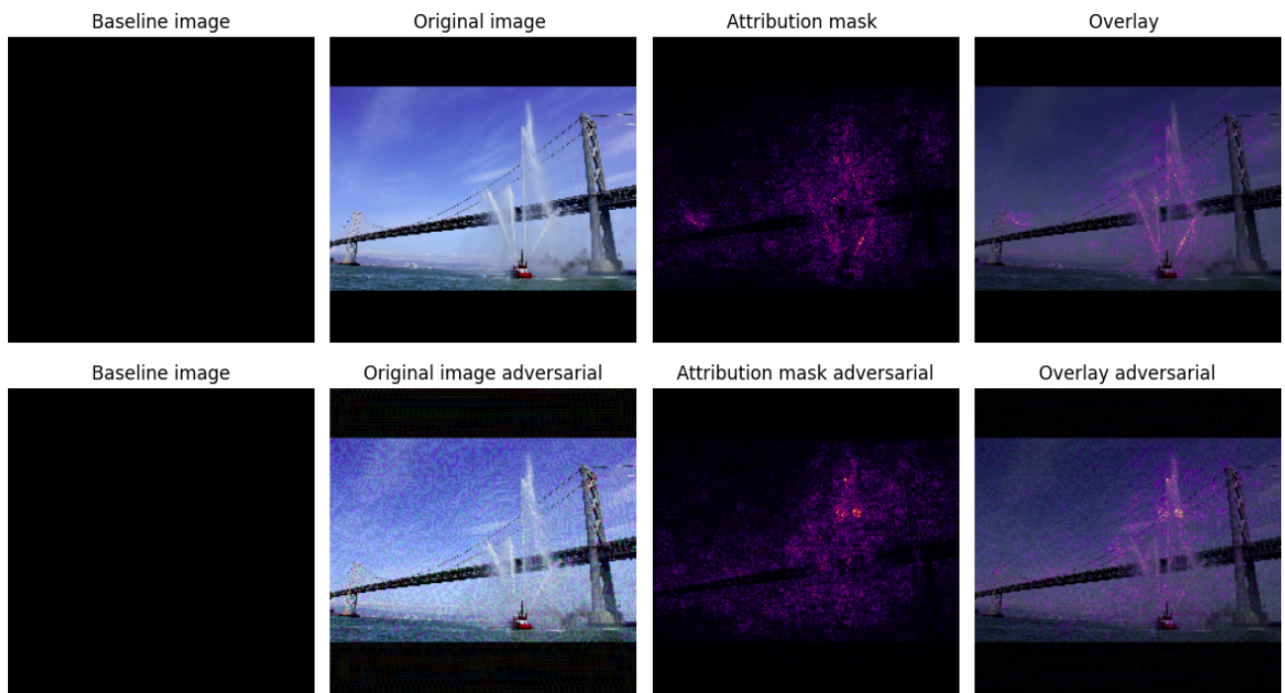


Figure 7: IG on original and adversarial Fireboat images

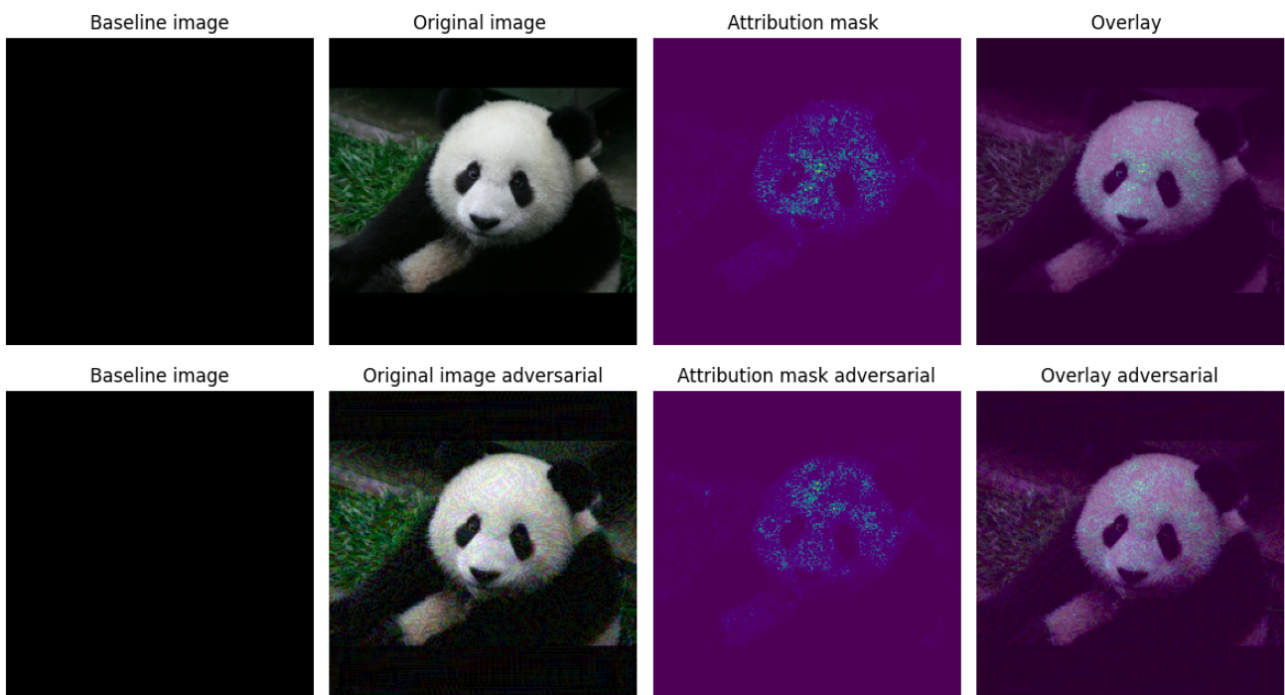


Figure 8: IG on original and adversarial Giant Panda images

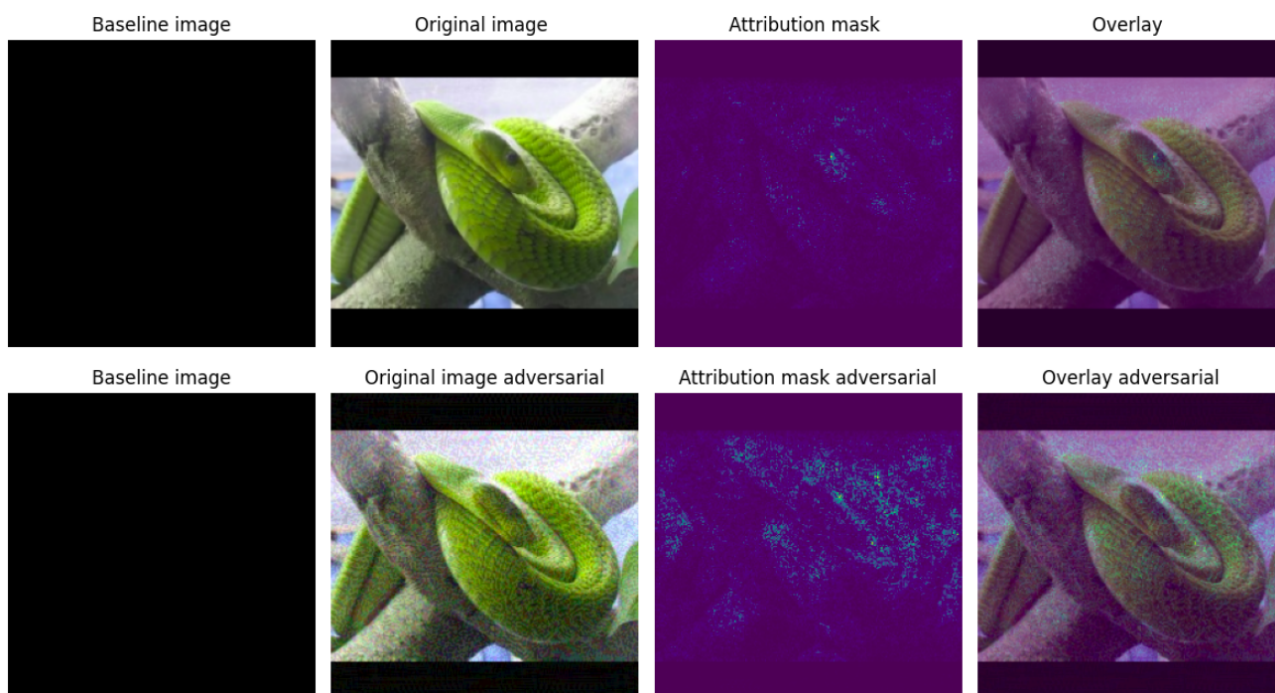


Figure 9: IG on original and adversarial Green Mamba images

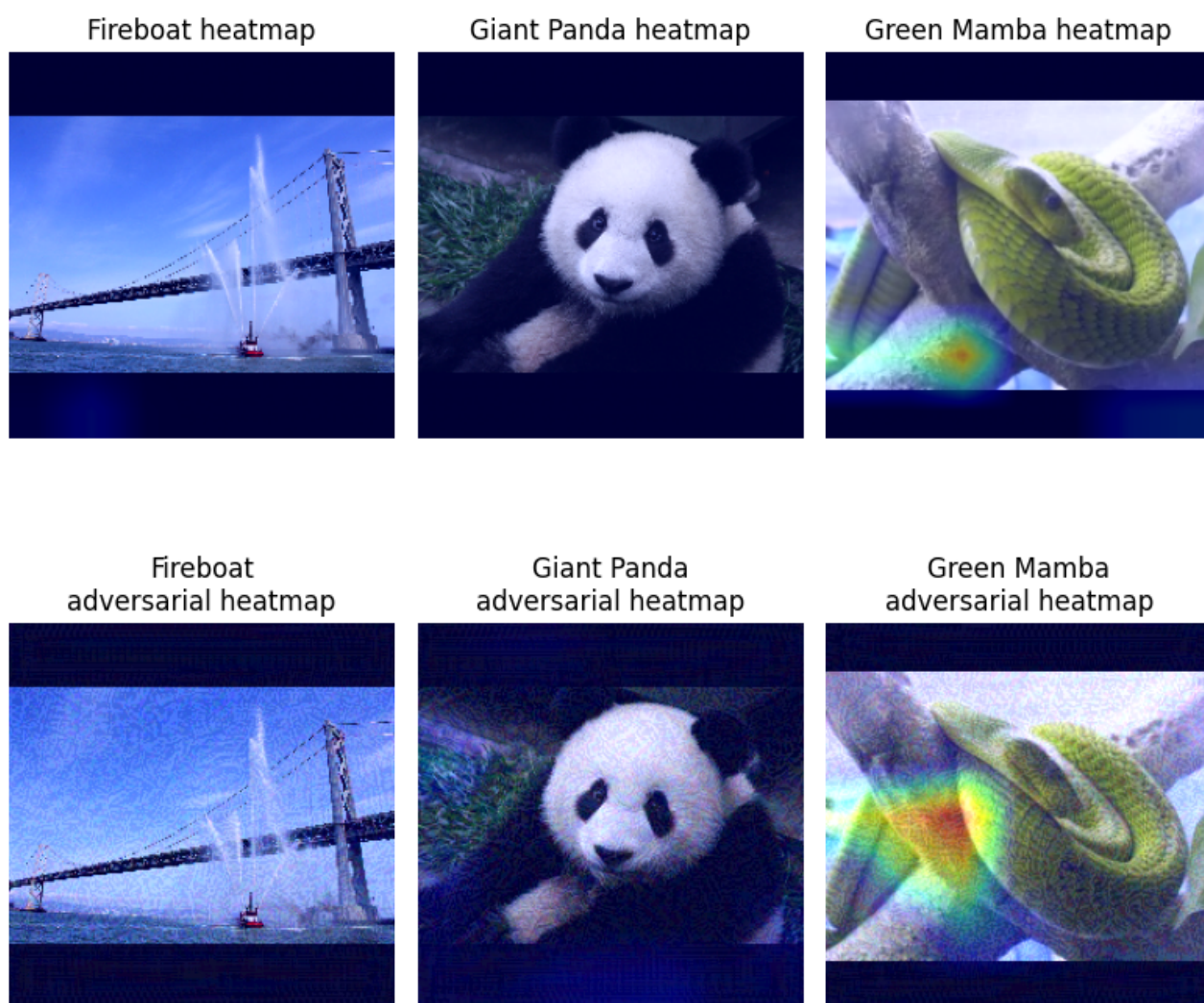


Figure 10: Grad-CAM on original and adversarial images

5 Effect of Using Softmax Layer for Grad-CAM and Pre-Softmax Gradients for IG

We also compared the impact of computing Grad-CAM on softmax probabilities and IG on pre-softmax gradients. Figures 11–14 show the results.

Observations:

- Highlighted pixels are reduced and more focused.
- Fireboat and Giant Panda: IG regions shrink to key object regions (bridge, panda body).
- Green Mamba: slight reduction in highlighted regions.
- Heatmaps show sharper focus on relevant features.

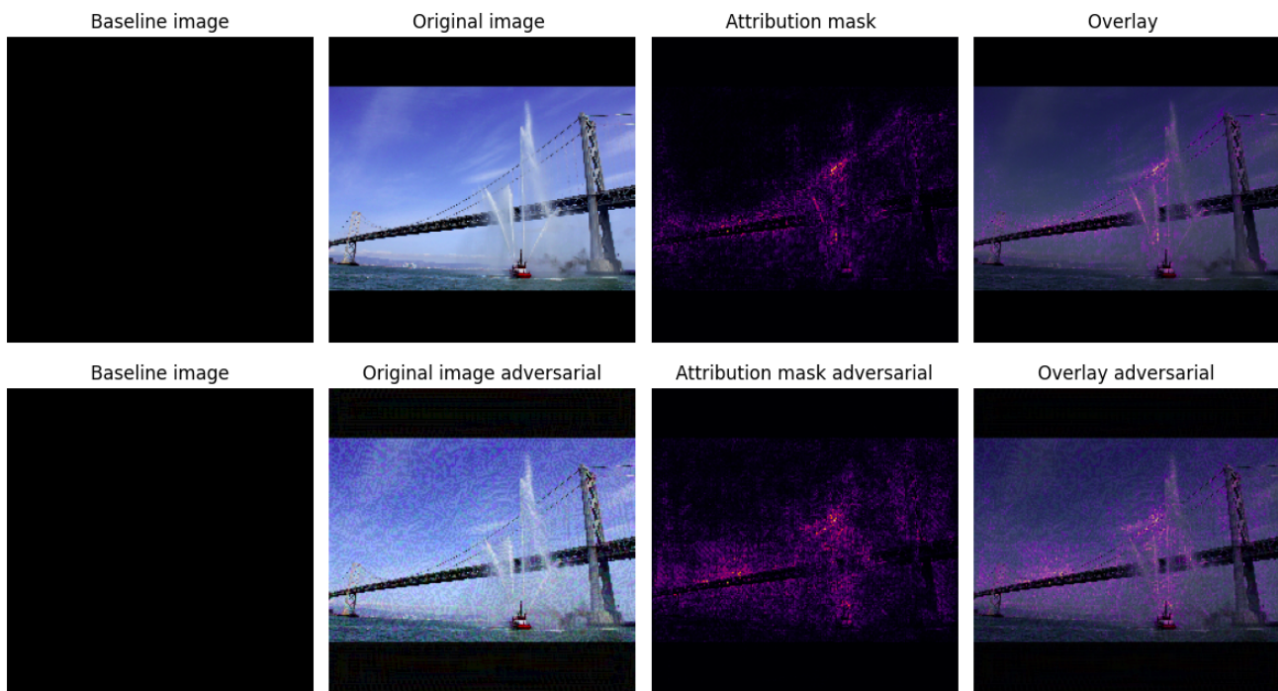


Figure 11: IG on Fireboat images using pre-softmax gradients

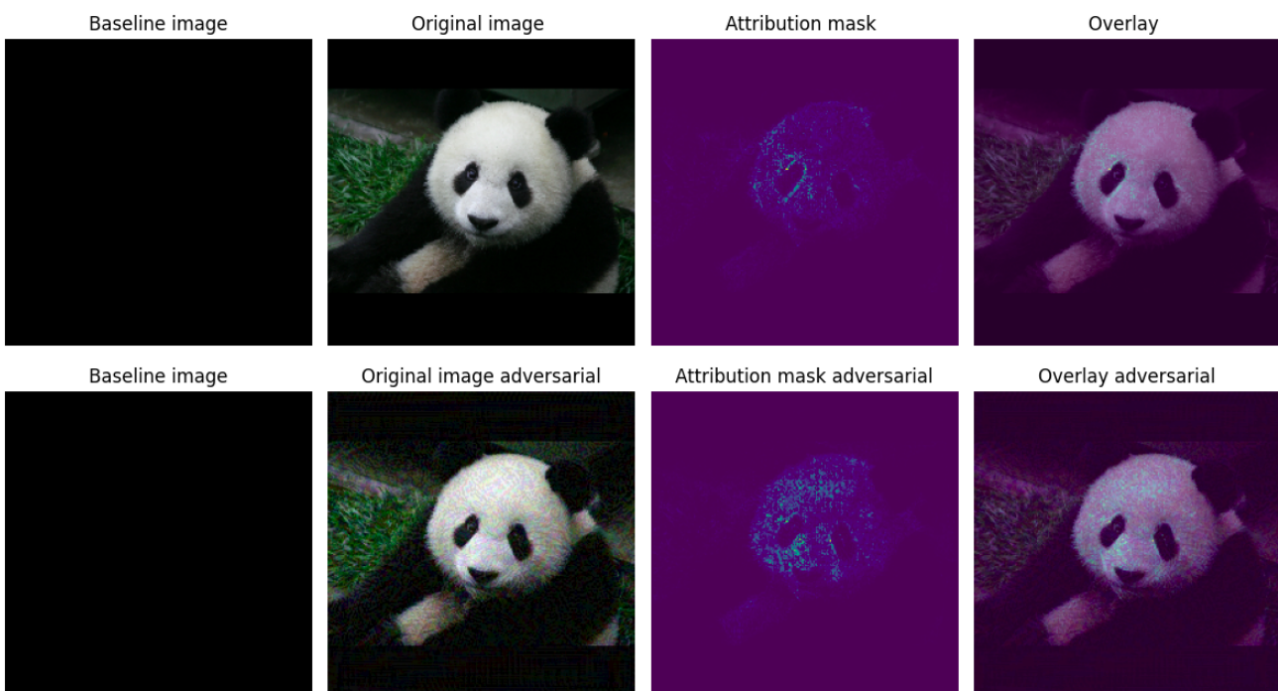


Figure 12: IG on Giant Panda images using pre-softmax gradients

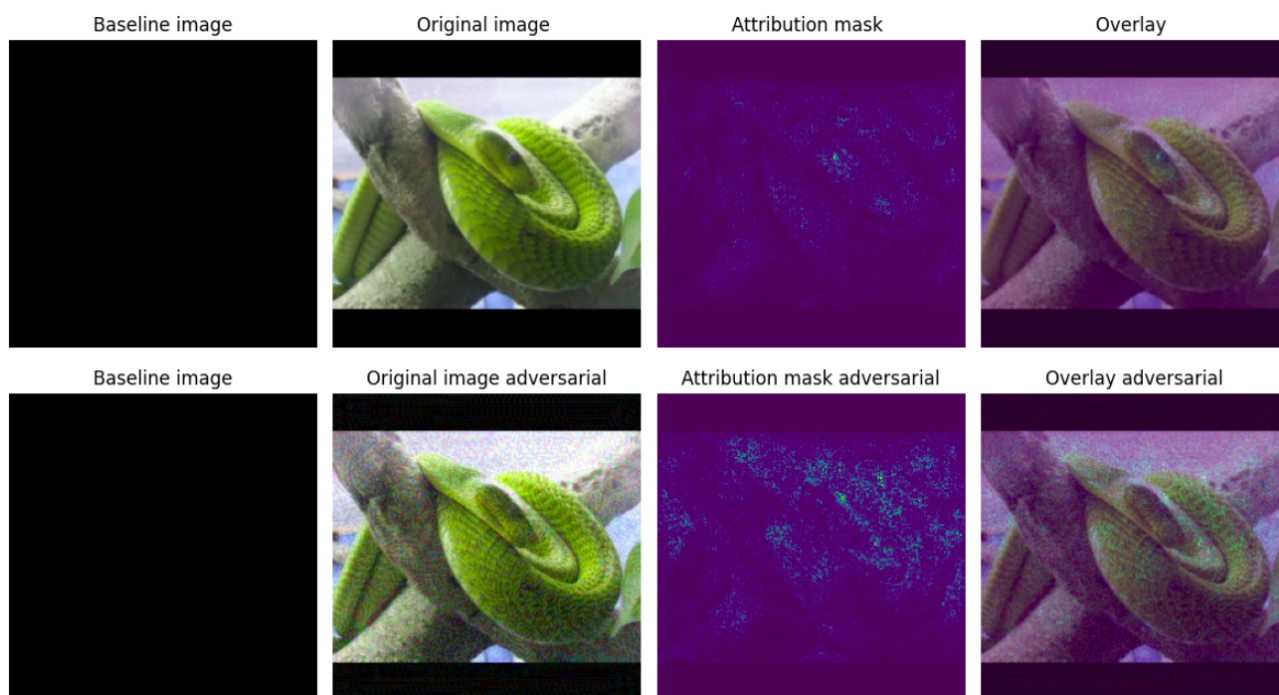


Figure 13: IG on Green Mamba images using pre-softmax gradients

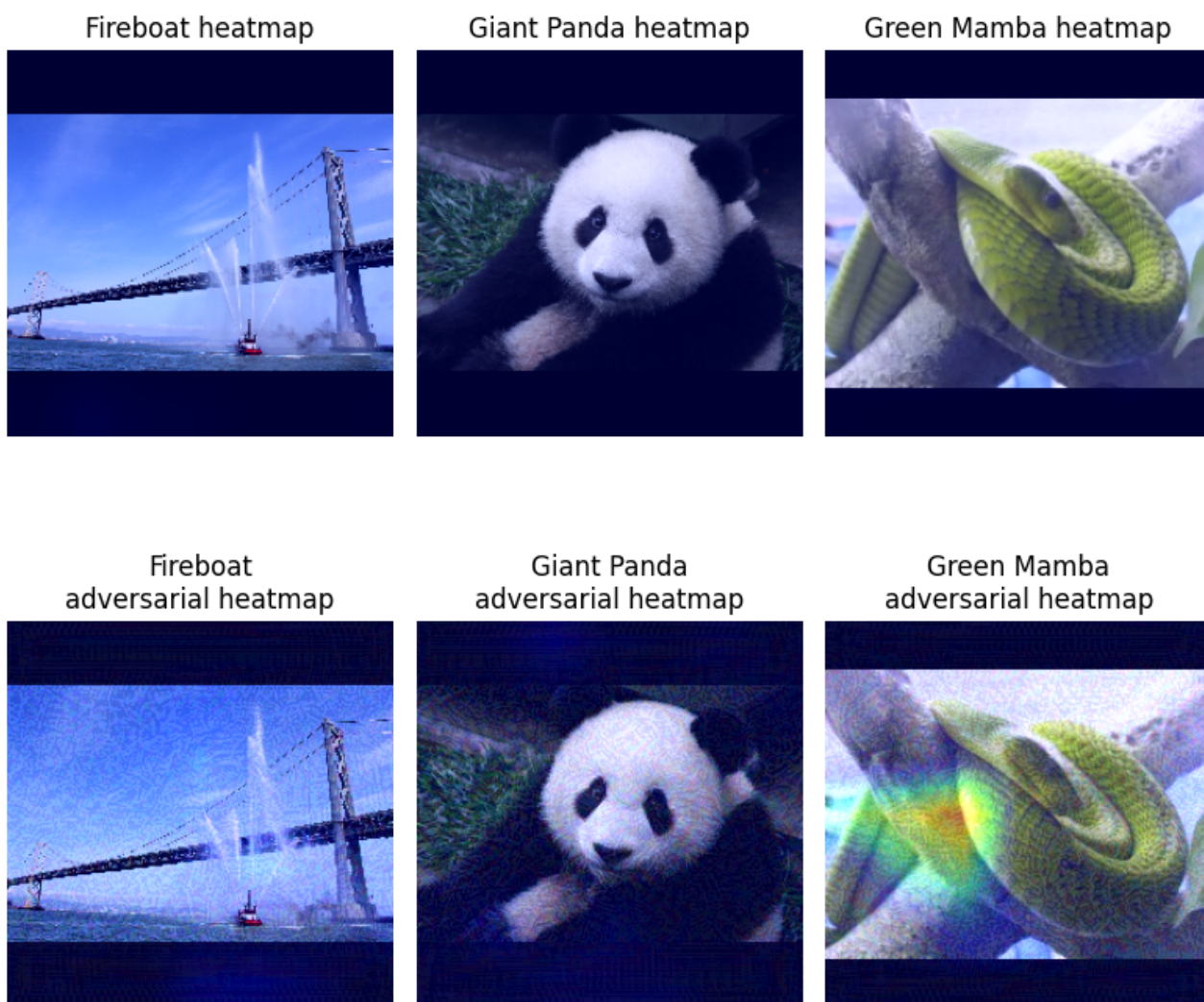


Figure 14: Grad-CAM on original and adversarial images using softmax

6 Results and Discussion

- Adversarial examples can significantly alter model predictions and spread important regions in attribution maps.
- IG and Grad-CAM highlight complementary regions responsible for model confusion.
- Using softmax for Grad-CAM and pre-softmax for IG produces more focused saliency maps, potentially improving interpretability.
- Fireboat and Green Mamba are more sensitive to adversarial perturbations than Giant Panda in this experiment.

7 Code

[Github code link 6.1 & 6.2](#)

[Github code link 6.3](#)